

Technical Audit Memo: Money Growth and Inflation

Objective: This is an empirical setup to audit the quantity theory of money using a 160-country macro panel (1991–2024). The purpose is to decompose the broad money-inflation relationship into two distinct estimands: the long-run cross-country average and the short-run annual within-country pass-through. The core finding is that while the long-run cross-country relationship remains near one-for-one, the clean short-run within-country pass-through collapses to near-zero post-2008.

1. Data Engineering & Workflow

The construction of the unbalanced panel involved merging raw World Bank WDI variables and managing severe country-specific data irregularities. This project prioritizes descriptive reliability over sample size maximization. Key data governance decisions included:

- **Outlier Mitigation (The 40% Threshold):** I trimmed the primary analytical sample to exclude countries with any year of inflation exceeding 40%. This prevents hyperinflationary regimes (which mechanically exhibit a 1:1 money-inflation mapping) from dominating the aggregate slope and obscuring the dynamics of stable-inflation economies.
- **Panel Consistency ($n \geq 30$):** For long-run averages, I imposed an $n \geq 30$ threshold to ensure sufficient time-series depth for country-level aggregation.
- **Unit Break Correction:** The raw WDI data contained a spurious -680% M2 growth rate for Sierra Leone in 2015. Investigation revealed this was a reporting/redenomination artifact (the M2/GDP level series dropped from $\sim 13,000$ to ~ 15). This observation was explicitly scrubbed via standard dropna handling.

2. The Empirical Contrast

Why separate long-run versus short-run? Because the estimands capture fundamentally different economic mechanics. The long-run captures regime differences; the short-run captures annual transmission.

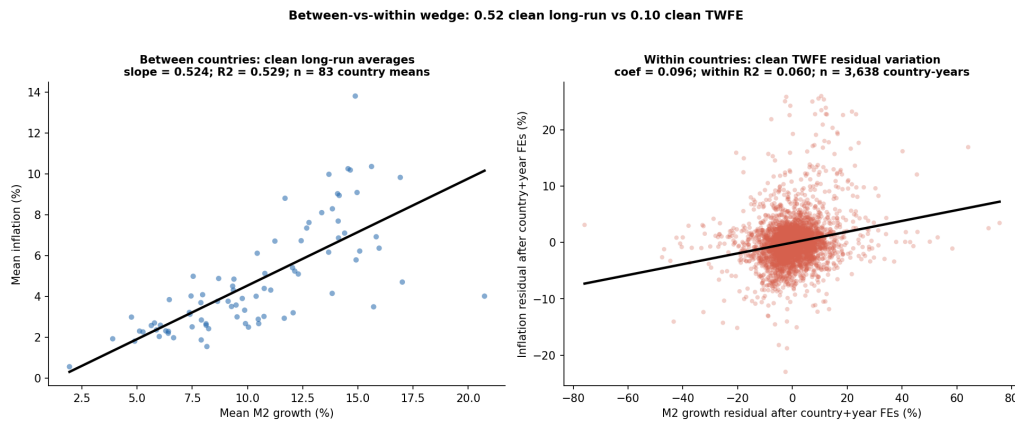
Estimand A: Long-Run Country Averages

$$\text{average_inflation}_i = \alpha + \beta \times \text{average_m2_growth}_i + \varepsilon_i$$

Countries with higher average M2 growth consistently have higher average inflation. Over 30 years, the full-sample slope is 0.952 ($R^2 = 0.877$). Even in the clean sample, the slope remains strong (0.524). The classic Lucas (1996) result survives.

Estimand B: Short-Run Within-Country TWFE

$$\text{inflation}_{it} = \gamma_i + \delta_t + \beta \times \text{m2_growth}_{it} + \varepsilon_{it}$$

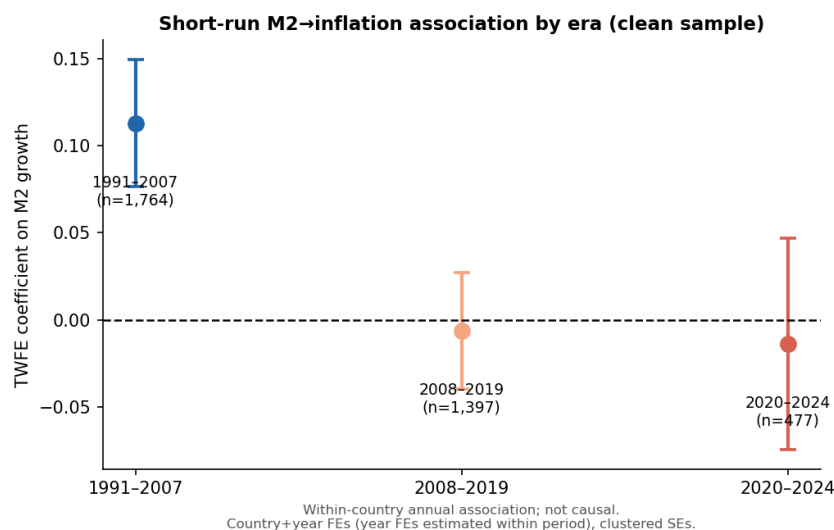


Within a country, do years with higher M2 growth see higher same-year inflation? The clean sample coefficient collapses to 0.096.

Why the pre/post-2008 split? The 2008 financial crisis fundamentally altered monetary transmission (e.g., QE, excess reserves). Splitting the TWFE exposes the structural shift:

- **1991–2007:** Slope is 0.113 ($p < 0.001$).
- **2008–2019:** Slope is -0.006 and statistically insignificant ($p = 0.711$).
- **2020–2024:** Slope is -0.014 and statistically insignificant ($p = 0.655$).

Short-run pass-through drops to near-zero in both clean-sample post-2008 cuts.



3. Key Limitations & Methodological Restraint

This project is strictly a descriptive empirical audit. I intentionally avoided causal overclaiming:

- **Instrumental Variables (IV) Rejected:** I explored Local Projections IV (LP-IV) using external lending rates and lagged M2 as instruments. The primary instrument yielded a first-stage F-statistic of 5.96. The secondary instrument was mechanically strong ($F = 16.29$) but the exclusion restriction was not credible. Therefore, I explicitly chose not to run headline IV. A weak identification strategy damages credibility more than admitting the limits of OLS. The IV results are retained solely as a directional appendix.

- **Inflation Targeting (IT):** I ran an event study on IT adoption. While IT adopters show a sharp post-adoption drop in the money-inflation slope (from 0.279 to 0.038 in the clean sample), IT adoption is endogenous. This is associational heterogeneity, not a causal treatment effect.
- **COVID as Stress Test, not Design:** The 2020–2023 COVID episode shows a positive cross-country money-inflation slope (0.494), but it cannot isolate broad money from fiscal transfers or supply shocks. It remains descriptive.